

MIHIR RAVINDRA ADELKAR

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Software Engineer with 3+ years designing distributed systems from first principles. Led ground-up microservices migration across 20+ engineers, 12 services, 7x faster api, 45% infra cost reduction. Shipped multi-agent LLM with MCP orchestration in production, replacing 3-day manual reviews with 5-second decisions backed by confidence scores and audit trails.

EDUCATION

Northeastern University

Master of Science in Software Engineering Systems

Boston, MA

Sept 2023 – May 2025

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, JavaScript, Java, SQL, Shell, HTML, CSS, SASS

Frameworks: FastAPI, React, Next.js, Node.js, Vite, Django, Spring Boot, Tailwind, MUI, GraphQL, gRPC

AI & LLM: OpenAI, MCP, RAG, Pinecone, Ollama, LangChain, LangGraph, Prompt Engineering, Agentic Workflows

Data & Infra: PostgreSQL, Redis, Kafka, MongoDB, Elasticsearch, AWS, GCP, Docker, Kubernetes, Terraform, Vercel

Tools: Prometheus, Grafana, Sentry, New Relic, Jest, Cypress, K6, Supabase, Claude Code

EXPERIENCE

Full-Stack Developer

Aug 2025 – Present

KeelWorks Foundation

Boston, MA

- Own end-to-end development of KeelCompass, an internal community discussion forum; build React + TypeScript, Tailwind UI with Node.js, PostgreSQL microservices, shipping features from spec to production on a small team
- Built Claude API-powered workflow automation for contextual recommendations and document generation; integrated LLM calls into Node.js services with prompt templates and structured output parsing
- Built LLM agent for community engagement content generation; designed RAG pipeline retrieving from existing blog corpus via Pinecone, LangChain-orchestrated workflows, MCP-managed context, and real-time event ingestion
- Built internal Claude-powered CLI to scaffold Jest and React Testing Library suites, generating 100+ test files across the team and removing hours of boilerplate per feature
- Automated end-to-end CI/CD deployment for containerized services on EKS with Docker and Terraform; used Claude Code for scaffolding and PR-quality reviews to ship same-day

Software Engineer

May 2023 – Sep. 2023

Freelancing (Dabadu.ai)

Remote

- Stabilized production services for Dabadu XRM, an automotive dealership CRM with AI-driven lead workflows; resolved defects across Socket.IO real-time chat, Twilio calling, S3 bulk upload, and Firebase push notifications
- Refactored legacy JS to TypeScript and introduced SonarCloud quality gates in CI, reducing production bugs 20%

Software Engineer

Jun. 2021 – May 2023

NeoSOFT Technologies

Mumbai, India

- Audited a 500K+ user real estate CRM; designed full microservices architecture with gRPC, Redis, Kafka boilerplate; coordinated 20+ engineers, 6 months, 12 services migrated, 7x faster inter-service calls, 45%+ infra cost reduction
- Designed GraphQL/REST APIs serving 500K+ users on PostgreSQL with Redis caching, cutting response times 70%; established Jest/Cypress automation reaching 85% test coverage
- Architected Django analytics engine aggregating real-time per-student classroom signals into a Next.js dashboard; enabled teachers to monitor live attendance, quiz scores, and engagement during class with post-session reports
- Engineered Homepage Builder CMS and multi-tenant CRM for Nesto using React, Node.js, and AWS Lambda; enabled non-engineers to update pages and promotions, removing a recurring engineering bottleneck
- Built CI/CD with mandatory tests and New Relic APM dual dashboards (engineering metrics + stakeholder health view), cutting MTTR from 25 to 15 minutes
- Built open-source dev tooling — mf-shell (Webpack Module Federation micro-frontend boilerplate) and Console (multi-tenant admin panel with RBAC); adopted as org-wide standards by 15+ internal teams

RECENT PROJECTS

Healthcare Prior Authorization AI | *Python, FastAPI, Claude API, MCP, Pinecone, FHIR R4, Streamlit*

- Designed Model Context Protocol (MCP) orchestration layer for multi-agent LLM reasoning, dynamically composes clinical data and policy context per request, outputs decisions in 5–10 sec with confidence scores and full audit trails
- Built FastAPI backend exposing agent workflows via REST; clinical NLP pipeline maps unstructured notes to FHIR R4 resources at 95%+ extraction accuracy, replacing a 3-5 day manual review process
- Re-architected the orchestration layer in LangGraph: 5-node stateful graph with conditional routing to human review for low-confidence cases, demonstrating explicit-state workflow patterns alongside the MCP implementation

CVE Intelligence System | *Go, Python, Kafka, Pinecone, Llama 3, AWS EKS, Terraform, Istio*

- Built distributed microservices for real-time CVE ingestion using Kafka stream processing; developed RAG pipeline with Pinecone and Llama 3 enabling engineers to query vulnerabilities in plain English instead of manual searches
- Deployed on AWS EKS with Terraform IaC and Istio service mesh; Prometheus/Grafana monitoring with automated alerting and Jenkins CI/CD pipeline